

The Thar Desert

The most densely populated desert in the world, named for its iconic sand ridges accumulated over the past 1.8 million years



▷ AT RISK

The Thar Desert provides a haven for the Egyptian vulture, endangered elsewhere partly due to habitat destruction.

The Thar, or Great Indian Desert, belongs to two nations. Pakistan lays claim to around 15 percent, but the greater part lies in India, mainly in Rajasthan.

Undulating dunes give the Thar Desert its name, derived from *thul*, the local word for “sand ridges.” The region is made up of metamorphic gneiss, sedimentary rocks, and alluvial deposits that, over about 1.8 million years, have been covered by windborne sand. Parts of the Thar consist of sandy plains, others of continually shifting dunes. Barren hills and vast saline lakebeds are also scattered throughout the desert.

Seasonal dry southwesterly winds blow from March to July, when temperatures rise to 122°F (50°C). From July to September, monsoons bring an average 4 in (100 mm) of rainfall to the western Thar and as much as 20 in (500 mm) to the east. This is enough to support sparse scrub and grasses, as well as drought-tolerant tree species such as acacia. Grasses provide food for mammals and birds, including the critically endangered great bustard.

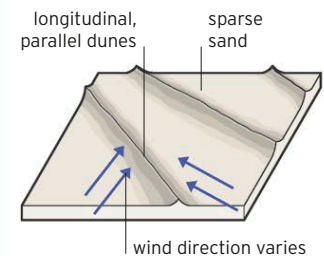


△ SEA OF SAND

Huge crescent-shaped dunes resembling rolling ocean waves cover the landscape of the arid western Thar.

HOW SEIF DUNES FORM

Most dunes are formed by multidirectional winds. When winds vary between two different, usually acutely angled, directions, longitudinal or “seif” dunes are created. These may extend for 60 miles (100 km) or more.



◁ ANCIENT POPLARS

The Taklamakan's relatively temperate climate does support some vegetation, including Euphrates poplars, which grow near the Tarim River and stabilize the sand dunes.

DUNE MIGRATION

Taklamakan dunes move up to 30 ft (9 m) per year due to prevailing northwesterly winds. Sand blown to the top of a dune falls to the other side, gradually shifting the position of the dune.

